

MTH313M End Semester Examination
Part A (May 4, 2026)

5:00 pm - 6:30 pm (+30 minutes for DAP students)

Maximum marks for Part A: 45

Minimum marks for Part A: 0

Instructions:

1. DO NOT attach any additional sheet to this page.
2. Write your answers in the designated place. If your answer is not legible, you shall not get credit.
3. If necessary, you may write your answer using a simple fraction.
4. All sub-questions carry **negative marks of same weight for an incorrect answer** and **0 marks for no attempt**. For example, answering a question of 2 marks wrongly will get -2 marks
5. Statements written outside designated spaces shall not be evaluated.
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7. You may use last pages of the answer-script (for part B) provided to you for rough work for part A. There will be no makes for rough work.

Name:

Roll No.:

Question 6.

Let $\{N_t\}_{t \geq 0}$ denote a Poisson process with rate $\lambda > 0$.

(a) (2 marks)

Write down the Kolmogorov's Forward Equation for this process.

(b) (2 marks)

Write down the Kolmogorov's backward Equation for this process.

(c) (1 mark)

Are the Kolmogorov's Forward and Backward equations identical for a Poisson process? Underline the correct answer.

Yes/No.

(Instruction: The sub-question (c) will only be graded if above two parts (a) and (b) have been answered.)

Question 7.

Consider a Birth and Death process $\{X_t\}_{t \geq 0}$ with infinitesimal birth parameters $\lambda_0 > 0, \lambda_1 > 0, \lambda_2 > 0, \dots$ and infinitesimal death parameters $\mu_0 = 0, \mu_1 > 0, \mu_2 > 0, \dots$.

(a) (3 marks)

Given that $X_t = i$ for some time $t \geq 0$ and integer $i > 0$, the conditional probability that a birth occurs before a death is

(b) (2 marks)

Given that $X_t = i$ for some time $t \geq 0$ and integer $i > 0$. Let T denote the waiting time for either a birth or a death to occur. Then, $\mathbb{E}T =$

Question 1. Consider an $M/M/1$ queueing process with arrival rate $\lambda = 1$ per unit of time and service distribution with rate $\mu = 2$ per unit of time. Assume that the system has infinite capacity.

- (a) (2 marks) In the steady-state of the system, the average number of customers in the system is
- (b) (2 mark) In the steady-state of the system, the average time spent by a customer in the system is unit
- (c) (2 marks) In the steady-state of the system, the average number of customers waiting in the queue is
- (d) (3 marks) In the steady-state of the system, the probability that an arriving customer will find the the server to be idle is

Question 2. Consider an $M/M/1$ queueing process with arrival rate $\lambda = 4$ per unit of time and exponential service distribution with rate $\mu = 2$ per unit of time. Assume that the system can accommodate upto 4 customers at a time. It is understood that customers arriving to find the queue full leave immediately and are not considered to be in the system.

- (a) (3 marks) In the steady-state of the system, the probability distribution $(p_0, p_1, p_2, p_3, p_4)$ of the number of customers in the system equals (write as a row vector)
- (b) (2 marks) In the steady-state of the system, the average number of customers in the system is
- (c) (2 marks) In the steady-state of the system, the average number of customers waiting in the queue is

Question 3. Consider an $M/M/4$ queueing process with arrival rate $\lambda = 4$ per unit of time and exponential service distribution with rate $\mu = 1$ per unit of time per server. Assume that the system can accommodate upto 6 customers at a time. It is understood that customers arriving to find the queue full leave immediately and are not considered to be in the system.

- (a) (1 mark) Think of this queueing process as a Birth and Death process. If there are two customers in the system at some time, the effective death rate of the system at this time is
- (b) (1 mark) Think of this queueing process as a Birth and Death process. If one customer is waiting in the queue at some time, the effective death rate of the system at this time is
- (c) (1 mark) In this queueing process, if one server becomes inoperable, can the system reach a steady state? Underline the correct answer.

Question 4. Let $\{B_t\}_{t \geq 0}$ denote a standard Brownian motion.

- (a) (3 marks) Write down the distribution of $\begin{pmatrix} 1 + B_3 \\ B_5 - B_2 \end{pmatrix}$
- (b) (2 marks) The sequence of random variables $\left\{\frac{B_n}{n} : n = 1, 2, \dots\right\}$ converges almost surely to a random variable X . Then X is
- (c) (2 marks) $\mathbb{E}|B_2|^3 =$
- (d) (1 mark) With probability 1, are the paths of $\{B_t\}_{t \in [0,1]}$ locally Hölder continuous of order $\frac{1}{3}$? Underline the correct answer.
- (e) (2 marks) Consider the process $\{X_t\}_{t \geq 0}$ defined by $X_t = \frac{1}{2}(B_{4t+1} - B_1)$. Is $\{X_t\}_{t \geq 0}$ a standard Brownian motion? Underline the correct answer.

Question 5. Let $X_t = \sum_{j=1}^{N_t} Y_j, t \geq 0$ be a Compound Poisson Process with rate $\lambda > 0$, where $Y_1, Y_2, \dots$ are i.i.d. $N(1, 3)$ random variables, independent of $\{N_t\}_{t \geq 0}$.

- (a) (1 mark) Does the process $\{X_t\}_{t \geq 0}$ have independent increments? Underline the correct answer.
- (b) (1 mark) Does the process $\{X_t\}_{t \geq 0}$ have stationary increments? Underline the correct answer.
- (c) (2 marks) $\mathbb{E}X_t =$
- (d) (1 mark) The variance of X_t is
- (e) (1 mark) For $u \in \mathbb{R}$, $\mathbb{E}e^{iuX_t} =$

Please turn over

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Question 6. Let $\{N_t\}_{t \geq 0}$ denote a Poisson process with rate $\lambda > 0$.

(a) (2 marks) Write down the Kolmogorov's Forward Equation for this process.

(b) (2 marks) Write down the Kolmogorov's backward Equation for this process.

(c) (1 mark) Are the Kolmogorov's Forward and Backward equations identical for a Poisson process? Underline the correct answer.

Yes/No.

(Instruction: The sub-question (c) will only be graded if above two parts (a) and (b) have been answered.)

Question 7. Consider a Birth and Death process $\{X_t\}_{t \geq 0}$ with infinitesimal birth parameters $\lambda_0 > 0, \lambda_1 > 0, \lambda_2 > 0, \dots$ and infinitesimal death parameters $\mu_0 = 0, \mu_1 > 0, \mu_2 > 0, \dots$.

(a) (3 marks) Given that $X_t = i$ for some time $t \geq 0$ and integer $i > 0$, the conditional probability that a birth occurs before a death is

(b) (2 marks) Given that $X_t = i$ for some time $t \geq 0$ and integer $i > 0$. Let T denote the waiting time for either a birth or a death to occur. Then,

$\mathbb{E}T =$

Question 1. Consider an $M/M/1$ queueing process with arrival rate $\lambda = 1$ per unit of time and service distribution with rate $\mu = 3$ per unit of time. Assume that the system has infinite capacity.

- (a) (2 marks) In the steady-state of the system, the average number of customers in the system is
- (b) (2 mark) In the steady-state of the system, the average time spent by a customer in the system is unit
- (c) (2 marks) In the steady-state of the system, the average number of customers waiting in the queue is
- (d) (3 marks) In the steady-state of the system, the probability that an arriving customer will find the the server to be idle is

Question 2. Consider an $M/M/1$ queueing process with arrival rate $\lambda = 3$ per unit of time and exponential service distribution with rate $\mu = 1$ per unit of time. Assume that the system can accommodate upto 4 customers at a time. It is understood that customers arriving to find the queue full leave immediately and are not considered to be in the system.

- (a) (3 marks) In the steady-state of the system, the probability distribution $(p_0, p_1, p_2, p_3, p_4)$ of the number of customers in the system equals (write as a row vector)
- (b) (2 marks) In the steady-state of the system, the average number of customers in the system is
- (c) (2 marks) In the steady-state of the system, the average number of customers waiting in the queue is

Question 3. Consider an $M/M/4$ queueing process with arrival rate $\lambda = 4$ per unit of time and exponential service distribution with rate $\mu = 1$ per unit of time per server. Assume that the system can accommodate upto 6 customers at a time. It is understood that customers arriving to find the queue full leave immediately and are not considered to be in the system.

- (a) (1 mark) Think of this queueing process as a Birth and Death process. If there are two customers in the system at some time, the effective death rate of the system at this time is
- (b) (1 mark) Think of this queueing process as a Birth and Death process. If one customer is waiting in the queue at some time, the effective death rate of the system at this time is
- (c) (1 mark) In this queueing process, if one server becomes inoperable, can the system reach a steady state? Underline the correct answer.

Question 4. Let $\{B_t\}_{t \geq 0}$ denote a standard Brownian motion.

- (a) (3 marks) Write down the distribution of $\begin{pmatrix} 1 + B_2 \\ B_4 - B_1 \end{pmatrix}$
- (b) (2 marks) The sequence of random variables $\left\{\frac{B_n}{n} : n = 1, 2, \dots\right\}$ converges almost surely to a random variable X . Then X is
- (c) (2 marks) $\mathbb{E}|B_3|^3 =$
- (d) (1 mark) With probability 1, are the paths of $\{B_t\}_{t \in [0,1]}$ locally Hölder continuous of order $\frac{1}{3}$? Underline the correct answer.
- (e) (2 marks) Consider the process $\{X_t\}_{t \geq 0}$ defined by $X_t = \frac{1}{2}(B_{4t+1} - B_1)$. Is $\{X_t\}_{t \geq 0}$ a standard Brownian motion? Underline the correct answer.

Question 5. Let $X_t = \sum_{j=1}^{N_t} Y_j, t \geq 0$ be a Compound Poisson Process with rate $\lambda > 0$, where $Y_1, Y_2, \dots$ are i.i.d. $N(1, 5)$ random variables, independent of $\{N_t\}_{t \geq 0}$.

- (a) (1 mark) Does the process $\{X_t\}_{t \geq 0}$ have independent increments? Underline the correct answer.
- (b) (1 mark) Does the process $\{X_t\}_{t \geq 0}$ have stationary increments? Underline the correct answer.
- (c) (2 marks) $\mathbb{E}X_t =$
- (d) (1 mark) The variance of X_t is
- (e) (1 mark) For $u \in \mathbb{R}$, $\mathbb{E}e^{iuX_t} =$

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Yes/No.

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(a) (3 marks)

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Given that $X_t = i$ for some time $t \geq 0$ and integer $i > 0$. Let T denote the waiting time for either a birth or a death to occur. Then, $\mathbb{E}T =$

Question 1. Consider an $M/M/1$ queueing process with arrival rate $\lambda = 2$ per unit of time and service distribution with rate $\mu = 3$ per unit of time. Assume that the system has infinite capacity.

- (a) (2 marks) In the steady-state of the system, the average number of customers in the system is
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Question 2. Consider an $M/M/1$ queueing process with arrival rate $\lambda = 4$ per unit of time and exponential service distribution with rate $\mu = 1$ per unit of time. Assume that the system can accommodate upto 4 customers at a time. It is understood that customers arriving to find the queue full leave immediately and are not considered to be in the system.

- (a) (3 marks) In the steady-state of the system, the probability distribution $(p_0, p_1, p_2, p_3, p_4)$ of the number of customers in the system equals (write as a row vector)
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Question 3. Consider an $M/M/4$ queueing process with arrival rate $\lambda = 4$ per unit of time and exponential service distribution with rate $\mu = 1$ per unit of time per server. Assume that the system can accommodate upto 6 customers at a time. It is understood that customers arriving to find the queue full leave immediately and are not considered to be in the system.

- (a) (1 mark) Think of this queueing process as a Birth and Death process. If there are two customers in the system at some time, the effective death rate of the system at this time is
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Question 5. Let $X_t = \sum_{j=1}^{N_t} Y_j, t \geq 0$ be a Compound Poisson Process with rate $\lambda > 0$, where $Y_1, Y_2, \dots$ are i.i.d. $N(2, 1)$ random variables, independent of $\{N_t\}_{t \geq 0}$.

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